



A novel hybrid model based on deep learning and error correction for crude oil futures prices forecast

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ABSTRACT

Energy is a crucial basis for ensuring people's life quality and advancing social and economic development. Whereas, the crude oil futures prices are influenced by several factors such as energy policies, which shows strong non-linearity and high volatility. Accurately predicting price changes of crude oil futures is a challenging task. In this study, a new hybrid model, namely VILIG based on the "decomposition-prediction-ensemble-error correction" framework, is constructed to predict the closing prices of West Texas Intermediate (WTI) and Brent crude oil futures prices. First, variational modal decomposition (VMD) is applied to decompose the original price dataset into several sub-models. Then, each sub-model is used as an input to the Long-short term memory network (LSTM) and an improved golden jackal optimization (IGJO) algorithm is proposed to optimize the hyperparameters of the LSTM. Subsequently, an improved fully ensemble empirical mode decomposition with adaptive noise (ICEEMDAN) is introduced to perform a secondary decomposition of the error sequence and gate recurrent unit (GRU) is used to predict the decomposed error sequences separately. Finally, the prediction outcomes are obtained by linear aggregation. In addition, based on the point prediction results, this study presents kernel density estimation (KDE) for interval estimation to perform uncertainty prediction. The empirical results show that the VILIG model outperforms all the currently available major models in terms of quantitative evaluation metrics. The model can provide a data base for decision making for crude oil futures investors and market regulators.

1. Introduction

The world economy is currently in a period of change and transition, and energy market prices are greatly correlated with political, economic, social, and market psychological factors. Fundamentally, economic fundamentals and market demand are the key factors affecting energy price changes (Hagen, 1994), but due to the scarcity and non-renewability of energy resources, coupled with the uneven distribution and consumption of resources in countries around the world, the development of energy price changes are highly randomness. In particular, along with the worldwide spread of the new coronavirus epidemic in 2020, the world economy came to a temporary standstill, the global transportation industry came to a halt, the manufacturing industry shut down, the demand for crude oil and other energy sources shrank sharply, and the development of the world economy once entered a "dark moment". The market panic spread, and even the May

futures price of WTI crude oil was negative. After 2021, with the vaccination of the new human crown and strong economic stimulus from governments, energy demand began to grow slowly and global energy prices continued to rise. In the second half of the year, due to repeated epidemics leading to slow recovery of the energy supply chain, Europe and the United States natural gas inventories collectively low, thus leading to a staggering increase in 2021 IPE UK natural gas. And Europe is mostly gas electricity, electricity prices soar after natural gas prices soar, resulting in many industrial enterprises in Europe production is greatly reduced or temporarily closed factories. Energy prices in other parts of the world have also risen significantly, and coal stocks in Asia have also been in short supply; in addition, the international oil prices have been moving upward at a high level, with a cumulative increase of more than 60% in 2021. Currently, energy conflicts and crises are still ongoing and developing globally.

Among the energy needs, crude oil, as the chip of industry and the

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"blood" of transportation industry, is the fundamental energy, chemical raw materials and strategic resources essential for socio-economic development, and plays the most important role in economic development. At present, oil consumption still accounts for the first place of global fossil energy consumption and nearly one third of all energy consumption (Company, 2020), and adverse changes in its price can seriously affect a country's economic development, social stability and state security (García-Franco et al., 2021). As an important energy product, crude oil not only has general merchandise properties, but also financial and political attributes, and its price fluctuations also have significant impact on the world economy, financial markets, and regional stability (Aloui et al., 2016). Simultaneously, the complex properties of crude oil assets determine that the international crude oil market is a typical complex nonlinear system, and its price fluctuations are the result of a series of intertwined and integrated fundamental and non-fundamental risk factors due to the combined effects of supply and demand in the crude oil market, the U.S. dollar exchange rate, speculative trading, geopolitical conflicts, natural disasters, and other elements. In the past two decades international crude oil prices have fluctuated dramatically, showing characteristics such as a shortening of the cycle of large surges and surges, and a reduction in the time of smooth transition: in 2001 the Internet bubble burst economic recession, oil prices fell 40%; in 2008 the global financial crisis, oil prices fell 75%; in May to June 2009, the global economy continued to recover, the Fed's quantitative easing led to a weak dollar, coupled with OPEC production remains low, funds continued to flow into the energy sector, up 36.5% in two months. 2014–2016 shale oil revolution, oil prices fell 76%; 2018 U.S.-China trade struggle, the U.S.-Iran conflict, OPEC increased production, oil prices fell 42%; 2020 global new crown epidemic outbreak, WTI crude oil prices fell from a high of \$63.27/barrel to -\$37.63/barrel, with a maximum drop of 159% in the interval. Since crude oil futures are subject to high market risk, it is important to design scientific methods to precisely forecast the direction and extent of crude oil price changes, so that microeconomic agents can identify and hedge price risks in the crude oil market and find profit opportunities.

The rest of this study is shown below. Section 2 is a literature review on crude oil futures price forecasting. In Section 3, Variational Mode Decomposition (VMD), long-short term memory network (LSTM), gated recurrent unit (GRU), improved golden jackal algorithm (IGJO), improved complete ensemble empirical mode decomposition with adaptive noise (ICEEMDAN) and the framework of the VILIG model are introduced. In Section 4, the empirical analysis is presented and the VILIG model is compared with other crude oil futures price forecasting models. Subsequently, various evaluation index such as MSE, MAPE and grey correlation degree are introduced to comprehensively evaluate the performance of the VILIG model. In Section 5, we further analyze and discuss the above results. Section 6 is the conclusion and outlook of this study.

2. Literature review

Crude oil futures price forecasting models are divided into two main categories.

- (1) Qualitative forecasting models. Qualitative forecasting refers to experts using their personal industry experience and analytical judgement to make judgements about the nature and extent of the future development of things based on available historical information, and then synthesising the views of all parties in a certain format as the main basis for forecasting the future. Qualitative prediction of crude oil futures is mainly based on the analysis of supply and demand conditions in the crude oil market, crude oil inventories, economic conditions, international situation, market participation, cost changes, and conditions in related markets (U.S. dollar, gold, stocks, etc.) to make reasonable expectations of future price changes. The common qualitative

forecasting methods are subjective probability method (Luís, 2001), Delphi method (Liu et al., 2022), leading index method (Van Dinh, 2018), etc. The qualitative forecasting method is simple and fast, with greater flexibility, time and effort saving, and small cost; however, it is constrained by the participants' experience and judgment ability, and also vulnerable to their subjective factors, and the forecasting results can only estimate a relatively reasonable price range, lacking the basis for accurate judgment.

- (2) Quantitative forecasting model. Quantitative forecasting is a type of method that uses certain mathematical models and methods to process and analyze the data based on the historical statistics available to reveal the regular links between the relevant variables to predict and infer future price trends.

The most commonly used quantitative models are traditional econometric models, decomposition and integration models, machine learning models and hybrid models.

Traditional econometric models are based on economic theory to construct estimating equations that can effectively identify the factors affecting crude oil prices and provide strong explanatory results. The main models include Ordinary Least Squares (OLS) (Yin and Yang, 2016), Nochange (Mu and Ye, 2015), Random Walk (RW) (Alquist and Kilian, 2010; Murat and Tokat, 2009), Autoregressive (AR) (Christiane and Lutz, 2012), Autoregressive Integrated Moving Average (ARIMA) (Meena et al., 2014; Wang and Wu, 2012), Vector Autoregression (VAR) (Drachal, 2021; Lyu et al., 2017), Vector Error Correct Model (VECM) (Javan and Vallejo, 2016; Gencer and Demiralay, 2013), Generalized AutoRegressive Conditional Heteroskedasticity (GARCH) (Lin et al., 2020; Bildirici and Ersin, 2013), Autoregressive moving average model (ARMA) (Karia et al., 2013), etc. For example, Afshin et al. (Javan and Vallejo, 2016) used net long position data for each trading category on the NYMEX to analyze the impact of each category on oil prices and volatility through vector error correction models and obtained meaningful analytical conclusions. Gencer et al. (Gencer and Demiralay, 2013) used monthly data to study the relationship between crude oil prices and sectoral returns of 18 sub-indices of Borsa Istanbul, and found short and long-term dynamic equilibrium relationships of the studied series through multivariate time series regression analysis using VAR and VECM methods. Bildirici et al. (Bildirici and Ersin, 2013) proposed an expansion of the linear GARCH, fractional-integral (FI)-GARCH, and asymmetric power (AP)-GARCH models with LSTAR-type nonlinear modeling, and showed that partial integration of the GARCH family of models and asymmetric power improvements could provide better forecasting power for gasoline prices and also better capture the long memory and high volatility characteristics. However, it is difficult to describe the various complex nonlinear characteristics of oil price changes because the form of such models is set fixed and the application is premised on the assumption that the time series is linear or smooth.

The decomposition integral prediction method decomposes the nonlinear crude oil price time series into multiple characteristic model subsequences with relatively single frequency component and a residual sequence, and then forecasts and integrates them respectively to obtain the predicted value of crude oil price. Common decomposition techniques include Wavelet Analysis (Hamid and Shabri, 2017), Empirical Mode Decomposition (EMD) (He et al., 2016), Ensemble Empirical Mode Decomposition (EEMD) (Xu et al., 2020), Singular Spectrum Analysis (SSA) (Wang and Li, 2018), VMD (Jianwei et al., 2019), etc. Since price time series often contain abrupt changes and noise peaks, the wave spectrum analysis method has the ability to identify and remove such anomalous information from the time series. Therefore, it has certain advantages in predicting future price trends. Among them, Hamid et al. (Hamid and Shabri, 2017) used wavelet theory to decompose the daily WTI crude oil price series into sub-time series of different scales and selected the best decomposition components as inputs to a multiple linear regression model, which proved to have good forecasting results.

He et al. (2016) proposed a model based on multivariate empirical model decomposition and used the heterogeneity feature of price changes to model and forecast the crude oil market price development. Xu et al. (2020) used integrated EEMD technique to decompose the data into different IMFs and reconstruct the IMFs so that the cumulative error is minimized, using Brent and WTI datasets and comparing with ARIMA and artificial neural network models with improved prediction accuracy and reduced computational complexity. Although modal decomposition can better discriminate mutations and noise in the series, individual modal decomposition models can interfere with the prediction error at the decomposition level due to data frequency constraints. Currently, the instability of national financial and monetary policies and sudden changes in the international regional situation can have a significant impact on oil prices, and the irregular price changes will be more frequent and complex, thus the traditional single decomposition integrated forecasting model is difficult to accurately predict price changes in the face of extreme event samples.

Machine learning models are the mainstream tools for big data processing applications and application analysis. Machine learning is the process of constructing a model through algorithms and using data to train the model, and if the evaluated performance is lower than the set threshold, the model parameters are adjusted for optimization until the model output meets the accuracy requirements. Currently, machine learning methods and models represented by Artificial Neural Network (ANN) (Khazem and Mazouz, 2013), Support Vector Machine (SVM) (Zhao and Zeng, 2019; Zhou and Wu, 2012), Least Square Support Vector Regression (LSSVR) (Yu et al., 2017) and Genetic Algorithm (GA) (Fariba, 2011) are widely used in predicting crude oil prices. For example, Khazem et al. (Khazem and Mazouz, 2013) developed an artificial neural network model and used daily crude oil futures price time series data listed on the New York Mercantile Exchange Group (NYMEXG) from 1996 to 2006 to predict future prices. The results showed that the ANN model improved the prediction accuracy of the nonlinear crude oil price time series. Zhao et al. (Zhao and Zeng, 2019) proposed an SVM-based approach to study the timeliness of oil price reactions to news. In the face of breaking news, the support vector machine model can predict the trend development of oil prices under different time scales more accurately and timely. Lean et al. (Yu et al., 2017) used the LSSVR model with parameter uncertainty to predict the WTI crude oil spot price by first designing the probability distribution of user-defined parameters in the LSSVR and generating random parameters and, obtaining the LSSVR members and weighting the output after predicting them one by one, the results showed a significant improvement in its prediction accuracy and computational efficiency.

As a machine learning algorithm of artificial intelligence, the model is repeatedly learned and trained using oil price time series data, and the forecast accuracy of machine learning models is often better, but the interpretability of the prediction results obtained by such models is poor, and the machine learning algorithms are prone to the trap of overfitting and local optimal solutions (Chen et al., 2023). Therefore, in recent years, research has gradually started to cross-mix traditional econometric methods, decomposition integration methods and machine learning models to obtain more accurate prediction results by taking advantage of the advantages of each method. For example, Wei et al. (Xu et al., 2015) developed an integrated model for crude oil price forecasting based on SVM and stochastic time-dependent pattern modeling and recognition system (STEPMRS). In this model, the crude oil price trend is first modeled using vector machines, and then the error is predicted using STEPMRS. Finally, the forecast results of VECM and STEPMRS are integrated using support vector machines to make the final forecast values more accurate. Zhang et al. (2015) first applied the EEMD algorithm to decompose the crude oil price into several IMF components. Subsequently, the LSSVM model is introduced to forecast each IMF component. Then, the PSO algorithm is used to optimize the hyper-parameter of the LSSVM model.

Ding et al. (Ding, 2018) proposed a new decomposition integration

forecasting process (EEMD-ANN-ADD) that combines integrated EEMD and ANN. Initially, EEMD is introduced to decompose the crude oil series data. Then, ANN is utilized to forecast the decomposed individuals, and finally the final forecast value is constructed by integrating the combination, which results in a significant improvement in forecasting accuracy compared to a single model. Wu et al. (2019) used an improved EEMD and LSTM network model for forecasting and concluded that the model has promising applications in crude oil price forecasting. Zhang et al. (Zhang and Wang, 2021) developed an EEMD-PSO-LSSVM-GARCH model to respectively forecast the prices of crude oil futures. The results show that this hybrid method has good adaptability to changes in data sample structure and frequency, and the prediction accuracy is better than common prediction models. Liu et al. (2021) proposed a VMD-ANN method for forecasting futures prices. They used WTI crude oil prices as the study object. The outcome indicated that the proposed VMD-ANN model significantly outperformed other models in terms of numerical and trend accuracy. Wang et al. (Wang and Wang, 2019) proposed a novel hybrid neural network, DPFWR neural network, combining dual parallel feedforward neural network and wavelet analysis theory with stochastic time effective function to predict energy futures price time series, and the results proved that the model possesses high accuracy. However, the prediction of international crude oil prices still faces many uncertainties, especially in different samples or at different frequencies, and the prediction results can vary greatly. Therefore, scholars in international crude oil price forecasting need to find suitable forecasting models for specific situations or create dynamic models that can capture the trend of international crude oil prices, so as to better predict and fit the trend of international crude oil price fluctuations.

Although models based on the idea of "decomposition-prediction-ensemble" can significantly improve the accuracy of the model, after the preliminary decomposition, the upper IMF component still have a high degree of volatility. Hence, on the basis of the initial decomposition, scholars begin to perform a dual-stage decomposition on the sub-models that still have strong non-linearity after the initial decomposition to further extract the characteristics of the sub-models. Li et al. (2022) took the closing prices of WTI and Brent crude oil futures contracts as the subject of study. Initially, ICEEMDAN is introduced to decompose the price of crude oil futures into a number of IMF components. Subsequently, the SSCE of each IMF component are calculated. Then, the smallest SSCE in the IMF component is used for the secondary decomposition. Ultimately, each IMF component are predicted by KELM optimized by artificial gorilla troops optimization. Sun et al. (2022) constructed ICEEMDAN-PE-EMD-CSSA-KELM model to predict the WTI and Brent crude oil futures price. Firstly, ICEEMDAN is used to decompose the original price series into several sub-models. Subsequently, envelope entropy is used to calculate the complexity of each sub-model and reconstruct it into high-frequency, mid-frequency and low-frequency components. Subsequently, EMD is used to perform a dual-stage decomposition of the high-frequency component. Then, the kernel extreme learning machine optimized by chaotic sparrow search algorithm (CSSA-KELM) model is used to forecast the sub-models respectively. Ultimately, the CSSA-KELM model is used to non-linearly ensemble the predicted sub-models. Constructed ICEEMDAN-SSCE-TVMD-GTO-KELM model to predict the WTI and Brent futures oil price. As result, the R^2 of the ICEEMDAN-SSCE-TVMD-GTO-KELM model on the test set of WTI dataset is 0.9939. Zhang et al. (2021) proposed a VMD-RES-EEMD-PSO-KELM model to predict the Brent and WTI crude oil futures price. Initially, VMD is used to decompose the original price series into a number of sub-models, followed by EEMD to perform a secondary decomposition of the residual series in view of the strong complexity of the residual series. Ultimately, a particle swarm algorithm optimized KELM is introduced to predict each sub-model. The results indicated that on single-step prediction, the R^2 of the VMD-RES-EEMD-PSO-KELM model is 0.996, which is much higher than ELM and other models.

Nowadays, although there are a large number of price forecasting methods, there are several shortcomings.

- (1) Although a large number of studies have used intelligent optimization algorithms to optimize the hyper-parameters of deep learning models, intelligent optimization algorithms are prone to local optimization and slow convergence during the optimization process, which affects the final accuracy (Huang et al., 2023; Deng et al., 2020; Cai et al., 2021; Duan et al., 2022).
- (2) Although the model based on the "decomposition-prediction-ensemble" idea can significantly improve the prediction accuracy, the upper IMF component still has strong volatility and still has a large error after prediction, but most of the research neglects to deal with the error sequence (Zhou and Xu, 2023; Chen et al., 2021; Luo et al., 2018).
- (3) Most studies focus only on deterministic point forecasts, but crude oil futures prices are subject to many uncertainties as they are closely related to many factors, including energy policies and exchange rates. And deterministic point prediction cannot capture the upward and downward volatility profile of crude oil futures prices, which has been overlooked by scholars (Wu et al., 2022).

In this research, we constructed a new hybrid model, namely VILIG based on the "decomposition-prediction-ensemble-error correction" framework to forecast crude oil futures prices. Initially, we decompose the crude oil futures price into different IMF components using VMD, and then IGJO-optimized LSTM is proposed to forecast each IMF component individually, and ensemble the predicted values of each IMF component to obtain the preliminary prediction results of crude oil futures price. Subsequently, based on the fact that the preliminary prediction still has a large error, we introduce ICEEMDAN to perform a dual-stage decomposition of the error series and use GRU to predict each sub-model of the error series to make the prediction results more accurate. Ultimately, based on the model prediction errors, this study introduces kernel density estimation (KDE) to perform interval estimation.

The VILIG model proposed in this study has the following main innovations.

- (1) In this study, we introduce strategies such as tent mapping, cauchy variation to improve the golden jackal optimization algorithm (GJO). Then, the improved golden jackal optimization algorithm (IGJO) is used to optimize the hyper-parameters of the LSTM.
- (2) To further improve the robustness and accuracy of the model, the error series of crude oil futures prices are decomposed by ICEEMDAN. Then, GRU are utilized to forecast each IMF component.
- (3) The VILIG model is compared with 20 models including ICEEMDAN-PE-EMD-CSSA-KELM. The results indicated that the VILIG model outperforms the other models in various metrics.
- (4) Based on the errors arising from point prediction, we introduced KDE to perform interval estimation.

3. Materials and methods

The models and methods used in this paper are described in turn.

3.1. VMD

VMD is a new adaptive decomposition estimation method for non-stationary signals (Dragomiretskiy and Zosso, 2013). This method can tackle the problem of mode aliasing in the process of decomposing the original signal. EMD and EEMD decompose signals based on the recursive method, but mode aliasing is inevitable during decomposition, which affects the prediction effect. VMD processes the original signal

based on non-recursive method, and the obtained Intrinsic mode function (IMF) can have different features, thus avoiding mode aliasing and end effect, with better decomposition effect and higher robustness.

3.2. Improved golden jackal algorithm

3.2.1. Golden jackal algorithm (GJO)

The Golden Jackal Optimization Algorithm is a new meta-heuristic algorithm proposed in 2022 (Chopra and Ansari, 2022). The algorithm is a new intelligent optimization algorithm that mimics the cooperative hunting behavior of the golden jackal. The GJO algorithm consists of three basic steps, which are searching for prey, surrounding the prey and attacking the prey. The mathematical model is shown as follows.

(1) Search Space Model

The equation for the initialization of the golden jackal population is shown in Eq. (1).

$$Y_0 = Y_{\min} + rand \times (Y_{\max} - Y_{\min}) \quad (1)$$

Y_0 is the initial position of the golden jackal population. $rand$ is a random number between 0 and 1. $Y_{\max}$ and $Y_{\min}$ are the upper and lower bounds for solving the problem respectively.

In the GJO algorithm, the prey matrix is represented in Eq. (2).

$$Prey = \begin{bmatrix} Y_{1,1} \cdots Y_{1,j} \cdots Y_{1,d} \\ \vdots \\ Y_{i,1} \cdots Y_{i,j} \cdots Y_{i,d} \\ \vdots \\ Y_{n,1} \cdots Y_{n,j} \cdots Y_{n,d} \end{bmatrix} \quad (2)$$

Y_{ij} is the j -th dimensional position of the i -th prey. n is the number of prey. d is the dimensionality of the problem solution.

In the optimization process, the fitness function is used to estimate the fitness value of each prey. The matrix of fitness values for all prey is represented as follows in Eq. (3).

$$F_{OA} = \begin{bmatrix} f(Y_{1,1}; Y_{1,2}; \cdots; Y_{1,d}) \\ f(Y_{2,1}; Y_{2,2}; \cdots; Y_{2,d}) \\ \vdots \\ f(Y_{n,1}; Y_{n,2}; \cdots; Y_{n,d}) \end{bmatrix} \quad (3)$$

F_{OA} is the matrix of fitness values of the prey. $f(x)$ is the fitness function or objective function. The one with the best fitness value is the male jackal and the one with the second best fitness value is the female jackal.

(2) Search for prey models (exploration phase)

As is the nature of jackals, they know how to sense and follow their prey, but occasionally the prey will not be easily caught and thus escape. Therefore, jackals wait and look for other prey. Hunting is led by the male jackal in Eq. (4). Female jackals follow the male jackals in Eq. (5).

$$Y_1(t) = Y_M(t) - E \cdot |Y_M(t) - rl \cdot Prey(t)| \quad (4)$$

$$Y_2(t) = Y_{FM}(t) - E \cdot |Y_{FM}(t) - rl \cdot Prey(t)| \quad (5)$$

The t is the current number of iterations. $Prey(t)$ is the position of the prey at the t -th iteration. $Y_M(t)$ and $Y_{FM}(t)$ are the positions of the male and female jackals in the t -th iteration, respectively. $Y_1(t)$ and $Y_2(t)$ are the updated positions of the male and female jackals corresponding to the prey at the t -th iteration, respectively.

E is the escape energy of the prey. Eqs. (6)–(8) is shown below

$$E = E_1 * E_0 \quad (6)$$

$$E_0 = 2 * r - 1 \quad (7)$$

$$E_1 = c_1 * (1 - (t / T)) \quad (8)$$

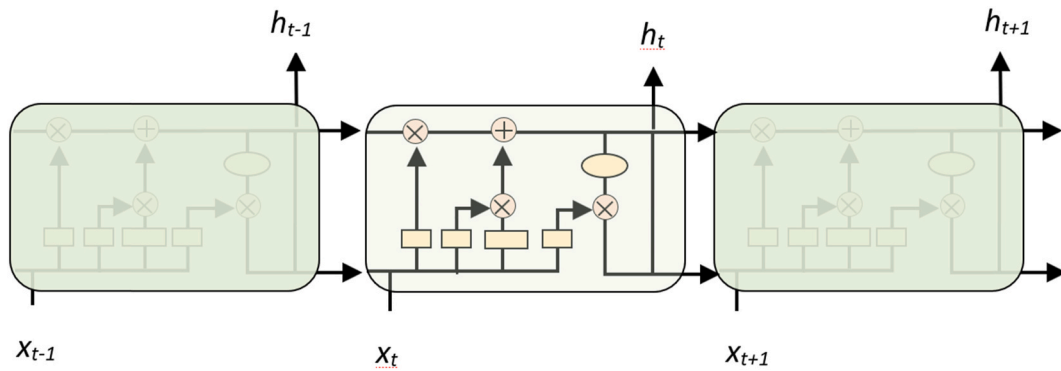


Fig. 1. Structure of LSTM

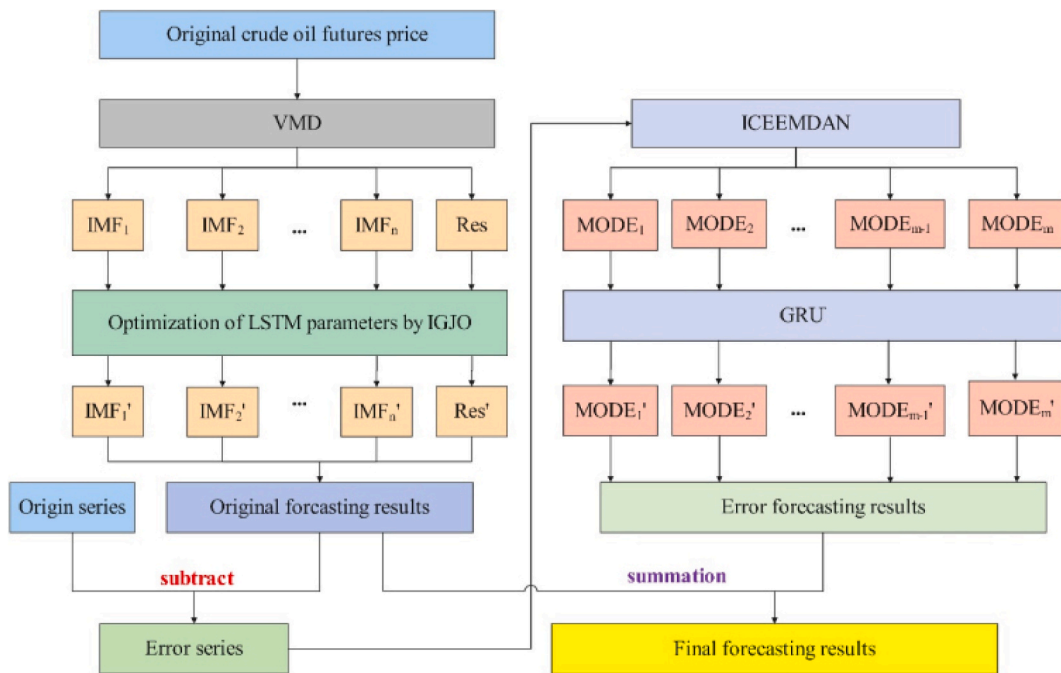


Fig. 2. VILIG model flow chart.

T is the maximum number of iterations. c_1 is a constant that takes the value 1.5. The t is the current iteration number. Throughout the iterations, E_1 decreases linearly from 1.5 to 0. In Eqs. (4) and (5), rl is a random number based on the Lévy distribution, which can be calculated by the following Eq. (9):

$$rl = 0.05 * LF(y) \quad (9)$$

$LF(y)$ is the Lévy flight equation as in Eq. (10).

$$LF(y) = 0.01 \times (\mu \times \sigma) / (|v^{(1/\beta)}|); \sigma = \left(\frac{\Gamma(1+\beta) \times \sin(\pi\beta/2)}{\Gamma(\frac{1+\beta}{2}) \times \beta \times (2^{\frac{\beta-1}{2}})} \right) \quad (10)$$

μ and v are a random number in the range of 0–1. β is a default constant that takes the value of 1.5.

In summary, the formula for updating the jackal's position is as follows in Eq. (11):

$$Y(t+1) = \frac{Y_1(t) + Y_2(t)}{2} \quad (11)$$

$Y(t+1)$ is the position of the jackal after the $(t+1)$ -th iteration.

(3) Surround and attack the prey (development phase)

When the prey is infested by jackals, their escape energy decreases and then the jackal pair will surround the prey detected in the previous stage. When the jackal pair encircles the prey, it starts to attack and devour the prey. The mathematical model is shown in Eqs. (12) and (13).

$$Y_1(t) = Y_M(t) - E \cdot |rl \cdot Y_M(t) - Prey(t)| \quad (12)$$

$$Y_2(t) = Y_{FM}(t) - E \cdot |rl \cdot Y_{FM}(t) - Prey(t)| \quad (13)$$

The t is the current number of iterations. $Prey(t)$ is the position of the prey at the t -th iteration. $Y_M(t)$ and $Y_{FM}(t)$ are the positions of the male and female jackals in the t -th iteration, respectively. $Y_1(t)$ and $Y_2(t)$ are the updated positions of the male and female jackals corresponding to the prey at the t -th iteration, respectively.

3.2.2. Improved golden jackal algorithm (IGJO)

Golden Jackal algorithm has the advantages of strong global optimization capability, which has extensively absorbed the attention of scholars and has been successfully applied to various fields. However, the Golden Jackal algorithm still has some shortcomings, such as poor

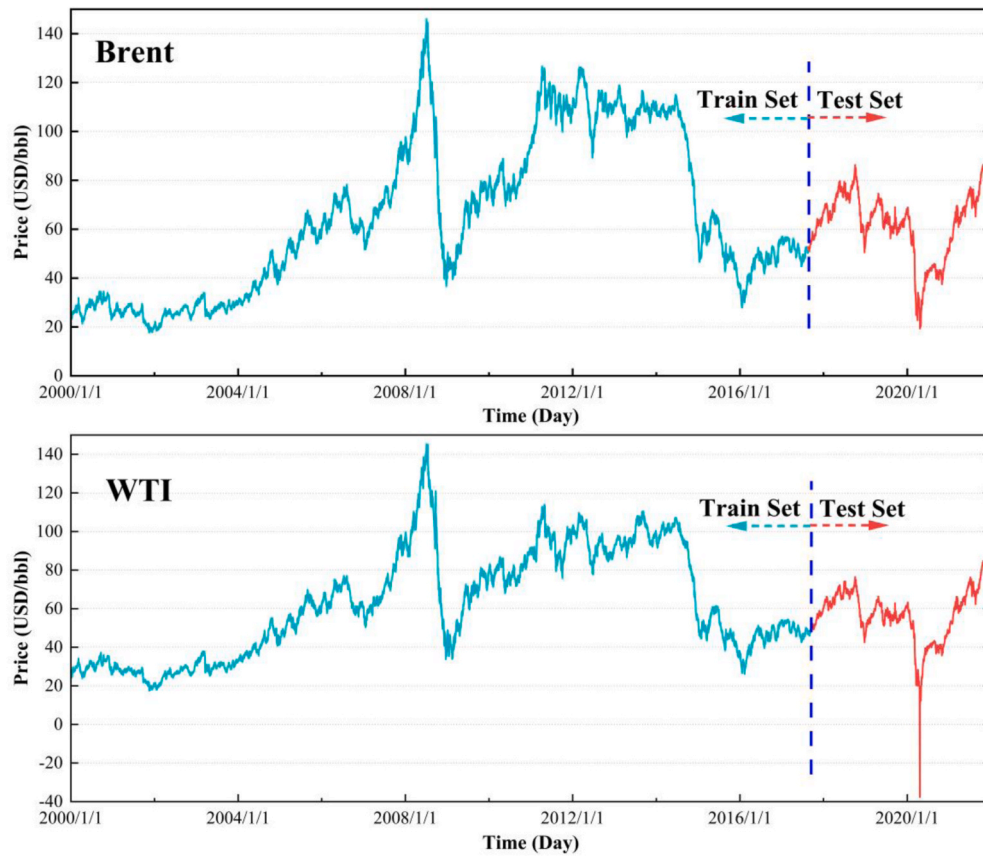


Fig. 3. Image of the data set.

Table 1

Descriptive analysis of the data set.

Crude Oil Futures Prices	Data Set	Count	Minimum value(USD/bbl)	Maximum value (USD/bbl)	Mean value (USD/bbl)	Standard deviation (USD/bbl) ²
WTI crude oil futures price	Total	5621	10.01	145.29	61.17	25.49
	Train Set	4498	17.45	145.29	62.13	27.67
	Test Set	1123	10.01	84.65	57.31	12.91
Brent crude oil futures price	Total	5648	17.68	146.08	64.49	29.09
	Train Set	4519	17.68	146.08	65.06	31.81
	Test Set	1129	19.33	86.40	62.21	13.31

local search ability, low convergence accuracy, slow convergence speed, etc. In this study, we introduces the following two strategies to improve the Golden Jackal algorithm.

(1) Tent mapping strategy

In the original golden jackal optimization algorithm, the population initialization is generated by a pseudo-random sequence, which leads to a very large uncertainty in the population diversity and does not guarantee that the solution solutions are uniformly distributed in the solution space. Therefore, it is necessary to improve the population initialization method. Due to the ergodic and random nature of chaotic mapping, if the chaotic mapping function is used to generate chaotic sequences as the positions of the initial population individuals, which will enhance the traversal uniformity of the initial solutions. In this study, we introduced the tent mapping to improve the initialization of the population. The formula of the Tent mapping is shown in Eq. (14):

$$x_{t+1} = \begin{cases} 2x_t, 0 \leq x_t \leq 0.5 \\ 2(1 - x_t), 0.5 \leq x_t \leq 1 \end{cases} \quad (14)$$

(2) Cauchy variation strategy

In the golden jackal algorithm, the golden jackal usually follows the current optimal value for global search, but due to its lack of good variation mechanism, the difference between individuals of the population will gradually decrease as the number of iterations increases, so the population aggregation phenomenon causes the algorithm to fall into local optimum and the solution accuracy is not high. In this research, we introduce the Cauchy variational operator to optimize the golden jackal algorithm. The formula of the Cauchy variational operator is shown in Eq. (15):

$$x'_i = x_i + \text{Cauchy}(0, 1) \times \text{rand} \quad (15)$$

$\text{Cauchy}(0, 1)$ is the standard Cauchy distribution. rand is a random number between 0 and 1. x_i is the currently selected Golden Jackal individual.

Since the Cauchy mutation operation will generate some new individuals, the population size is bound to exceed the original. Therefore, according to the competitive elimination mechanism, the individuals with the lowest ranking in each subspace are eliminated according to the fitness value when the algorithm completes each search, until the

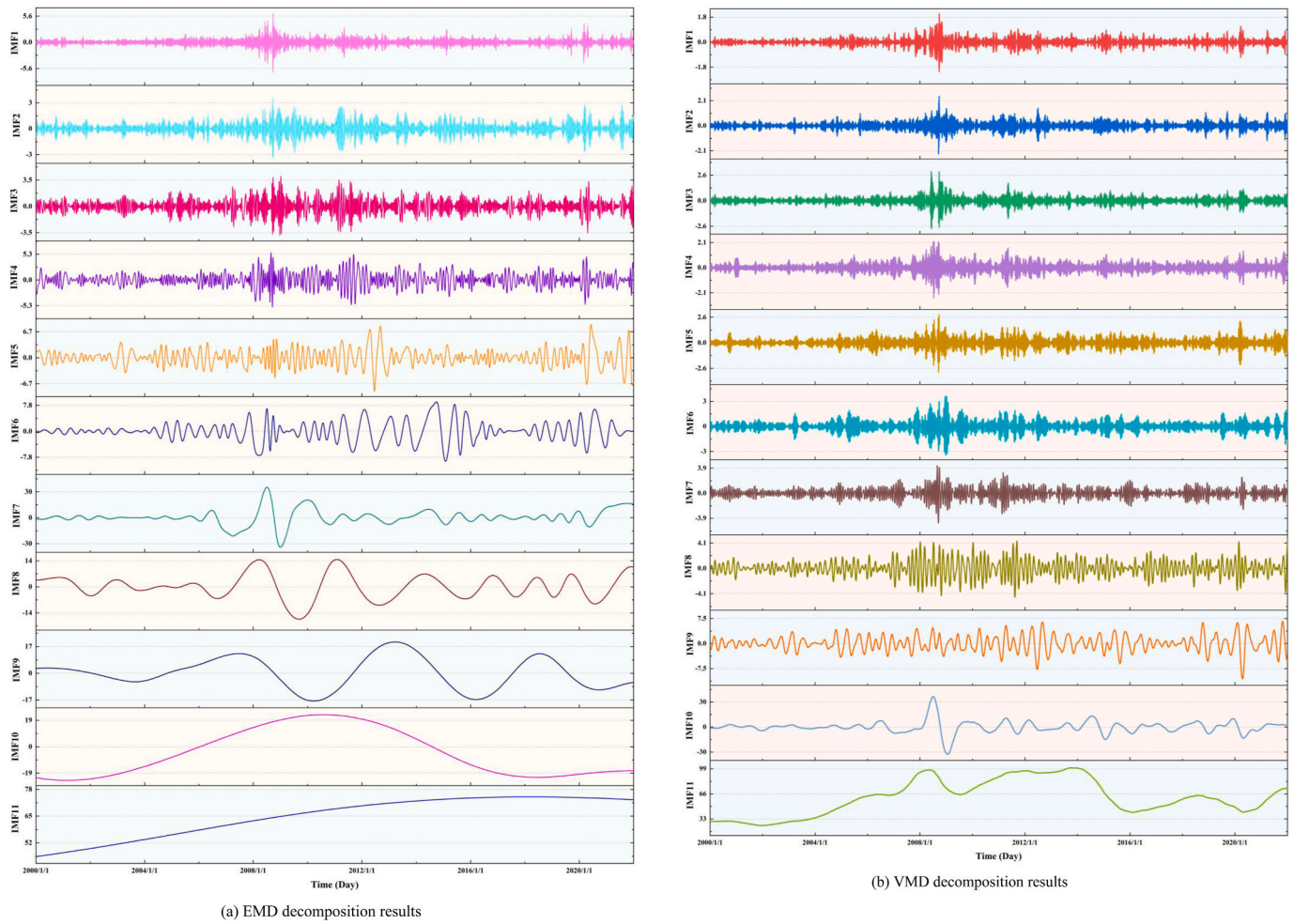


Fig. 4. EMD and VMD decomposition results for WTI daily crude oil futures prices.

population reaches the original size. In this way, the algorithm can be updated continuously after each search iteration, which can better improve the quality of the population individuals and ensure the quality of the solution.

3.3. LSTM

As an improved recurrent neural network, LSTM adds three gating units of forgetting gate, input gate and output gate to RNN, which validly tackle the problems of gradient disappearance or explosion defects in long sequences of RNN (Hochreiter and Schmidhuber, 1997). LSTM can effectively process long time sequences based on the unique gating units and memory units, overcoming the problem of long time dependence on input sequences. The forgetting gates retain important information while avoiding the backward transfer of useless information, while the input and output gates read the data and transfer the processed data to the next moment (Wu and Wang, 2022). With its superior performance, LSTM has been extensively introduced to various fields such as load prediction (Saeed et al., 2022; Qin et al., 2022), image classification (Yechuri and Ramadass, 2021; Zhou et al., 2019), etc. The structure of LSTM is shown in Fig. 1.

3.4. Improved complete ensemble empirical mode decomposition with adaptive noise (ICEEMDAN)

Colominas et al. proposed the ICEEMDAN method in 2014 to improve the CEEMDAN method (Colominas et al., 2014). The CEEMDAN method adds Gaussian white noise in the decomposition process,

while the ICEEMDAN method adds Gaussian white noise after the EMD decomposition has been performed. It mainly calculates the local mean of each modal component signal plus noise and defines the ICEEMDAN decomposition method improves the noise reduction effect by improving the situation of spurious components and modal blending in the early stage of decomposition reconstruction by traditional methods.

3.5. Gated recurrent unit

Although LSTM has been extensively utilized to many fields because of its excellent performance, due to the long training time, large number of parameters and complicated internal computation of LSTM, Cho et al. proposed a simpler GRU model by merging the cell states and hidden layer states of LSTM (Cho et al., 2014). Compared with LSTM, GRU has a simpler structure and fewer parameters, which makes it perfect not only for simulating problems with multiple input variables, but also for handling temporal prediction problems.

3.6. Hybrid model of crude oil futures prices forecasting

The VILIG model is divided into seven main parts as follows.

- (1) Pre-processing stage. VMD is suitable for forecasting highly volatile and non-stationary series, while the crude oil futures price series itself is characterized by highly volatile price changes and spatio-temporal non-stationarity. In this study, VMD is utilized to decompose the original crude oil futures price time series with strong stochasticity into a series of smooth modal components.

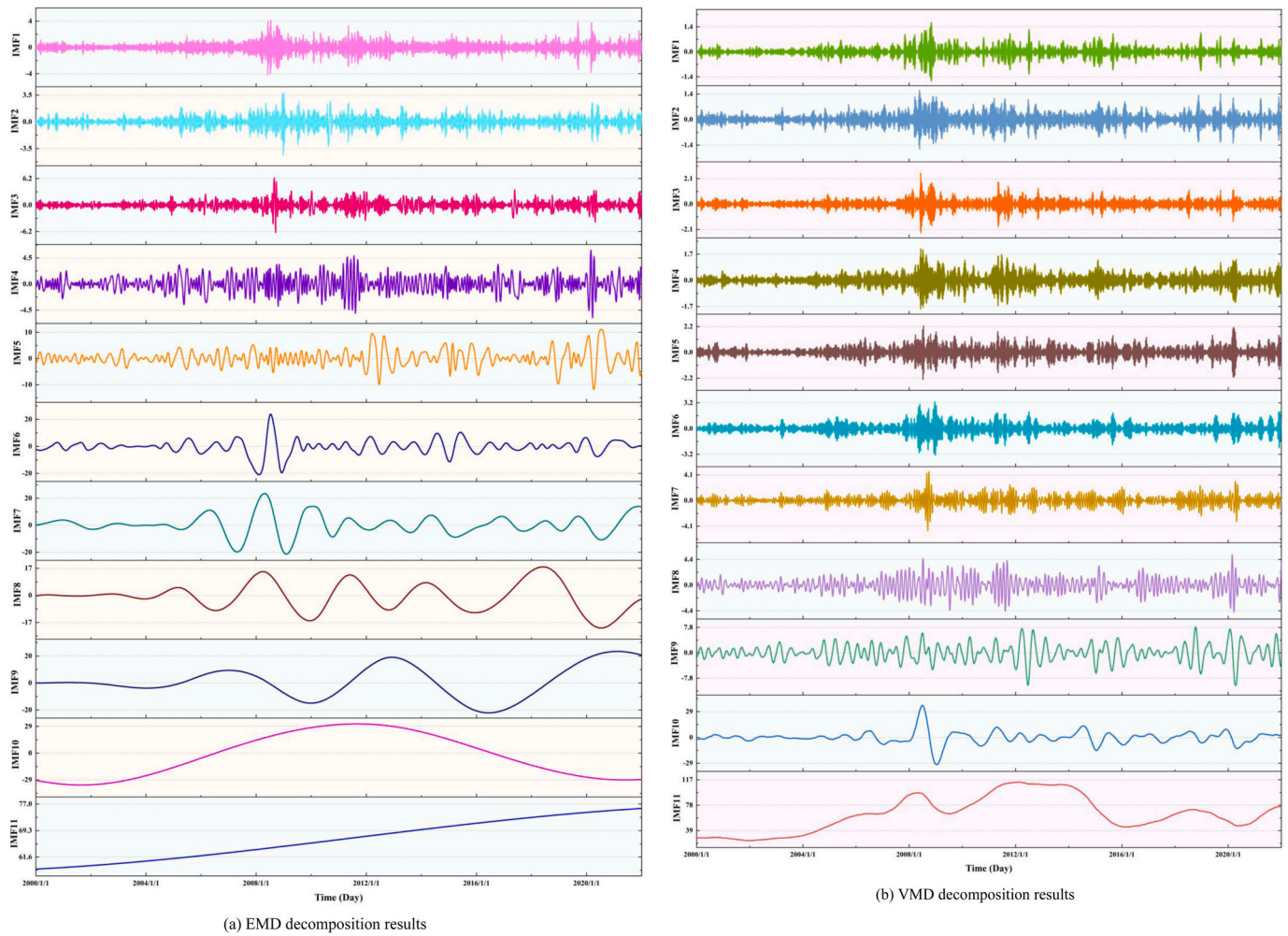


Fig. 5. EMD and VMD decomposition results for Brent daily crude oil futures prices.

Table 2
Model parameters.

Parameter	Value
The number of golden jackal population	5
The number of iterations	100
Dim	3
Learning rate	[0.001,1]
Maximum number of iterations	[10,500]
Number of neurons in the hidden layer	[10,300]

- (2) Parameter optimization of the IGJO-based LSTM. The LSTM is able to maintain memory of "information" and the long short-term information has different weights on the current price. This model feature is very suitable for forecasting oil futures market prices that have not yet reached full efficiency, especially in the futures market where recent information will have a higher influence on its price changes. Hence, we use the LSTM for forecasting for each IMF. Then, to overcome the subjectivity and arbitrariness in the selection of hyperparameters in the LSTM model, the hyperparameters are optimized using the Improved golden jackal algorithm to make them more objectively valid. We took the maximum number of iterations of the LSTM network as the optimization objects of the IGJO algorithm. Then, we initialized the parameters of the IGJO parameter.
- (3) A hybrid model was used to predict and evaluate each IMF component. The LSTM network model was constructed based on

- the optimized parameters by IGJO algorithm. After the model was trained, predictions were then performed on the test set.
- (4) The error series is obtained by subtracting the actual values from the predicted values. Based on the strong non-linearity and non-smoothness of the error series, the ICEEMDAN algorithm is applied to decompose the error series to reduce its non-smoothness because it established adaptive white noise component can decompose the error series more thoroughly and the information features are more obvious.
- (5) Since GRU has a more simplified structure and fewer parameters, we introduced GRU to predict each IMF component after the decomposition of the error series, and adjust the learning rate, maximum number of iterations and other related parameters to enhance the precision of the model.
- (6) The initial prediction series and the error prediction series are linearly summed to obtain the final prediction results. Subsequently, the mean squared error (MSE), root mean squared error (RMSE) are used to evaluate the performance of the model from several aspects.
- (7) Based on the errors in the point predictions, we introduce kernel density estimation to estimate the intervals separately at different confidence intervals.

The flow chat is shown in Fig. 2.

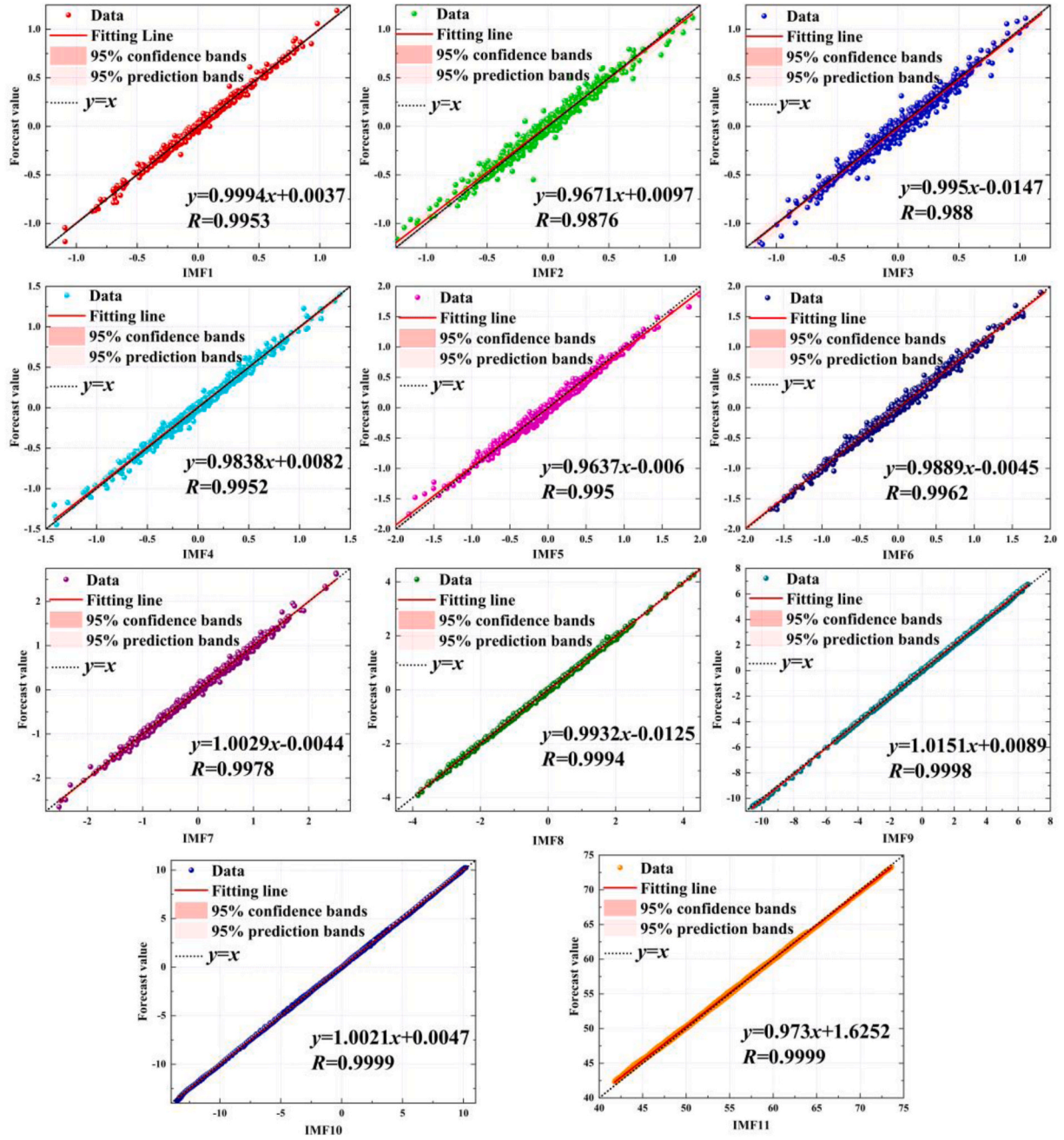


Fig. 6. Forecast results for each IMF component of WTI daily crude oil futures prices.

3.7. Error evaluation criteria

In this study, we chose four evaluation index to such as *MSE*, *RMSE* to assess the deterministic point performance of the VILIG model and other models. In addition, we chose the prediction interval coverage probability (*PICP*) and predicted interval average width (*PINAW*) to evaluate the performance of interval estimation of VILIG model. These evaluation indicators are calculated as shown in Eqs. 16–22.

$$MSE = \frac{1}{N} \sum_{t=1}^N (y_t - \bar{y}_t)^2 \quad (16)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{t=1}^N (y_t - \bar{y}_t)^2} \quad (17)$$

$$MAE = \frac{1}{N} \sum_{t=1}^N |y_t - \bar{y}_t| \quad (18)$$

$$MAPE = \frac{1}{N} \sum_{t=1}^N \left| \frac{y_t - \bar{y}_t}{y_t} \right| \quad (19)$$

$$R^2 = 1 - \frac{\sum_{t=1}^n (y_t - \hat{y}_t)^2}{\sum_{t=1}^n (y_t - \bar{y}_t)^2} \quad (20)$$

$$PICP = \left\{ \begin{array}{l} 1, \hat{y}_t \in [L_t, U_t] \\ 0, \hat{y}_t \notin [L_t, U_t] \end{array} \right\} \quad (21)$$

$$PINAW = \frac{1}{NR} \sum_{i=1}^N (U_i - L_i) \quad (22)$$

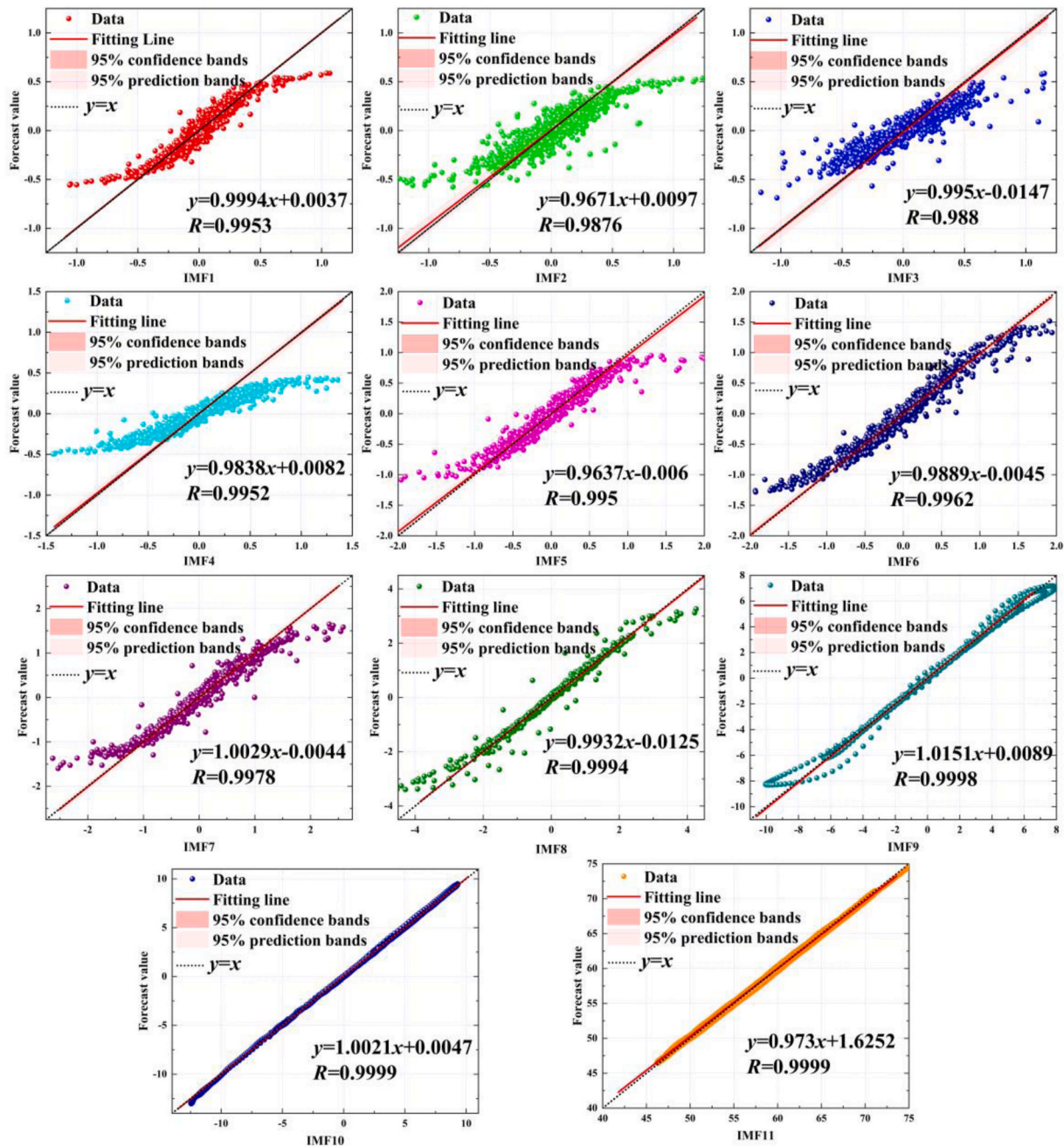


Fig. 7. Forecast results for each IMF component of Brent daily crude oil futures prices.

y_t is the actual value, $\bar{y}_t$ is the mean of the actual value and N is the number of test set. $RMSE$, MSE , mean absolute error (MAE) and mean absolute percentage error ($MAPE$) are all measures of prediction error. The smaller their values, the higher the prediction accuracy of the model. U_i and L_i are the upper and lower bounds of the prediction interval for sample i , respectively.

4. Case study

4.1. Data source and descriptive analysis

In this study, we selected crude oil futures prices for each trading day of WTI and London Brent from 01/04/2000 to 12/31/2021 as the object of study (Fig. 3). According to the principle of machine learning for data set splitting (Sun and Zhang, 2018), we set the first 80% of crude oil

futures prices for each trading day as the training set and the last 20% of crude oil futures prices for each trading day as the test set (i.e. we set the crude oil futures price from 01/04/2000–09/25/2017 as the training set and set the crude oil futures price from 09/26/2017–12/31/2021 as the test set).

Subsequently, we perform descriptive analysis on the datasets of daily crude oil futures prices for WTI and Brent, respectively. The results of the descriptive analysis are shown in Table 1.

4.2. VMD results

Among the parameters of the VMD, the determination of the number of decomposition layers k is crucial. In this study, to determine the number of decomposition layers of VMD. First, the raw data is automatically decomposed into multiple IMF components by EMD, and then the K value is determined based on the number of modes. This method

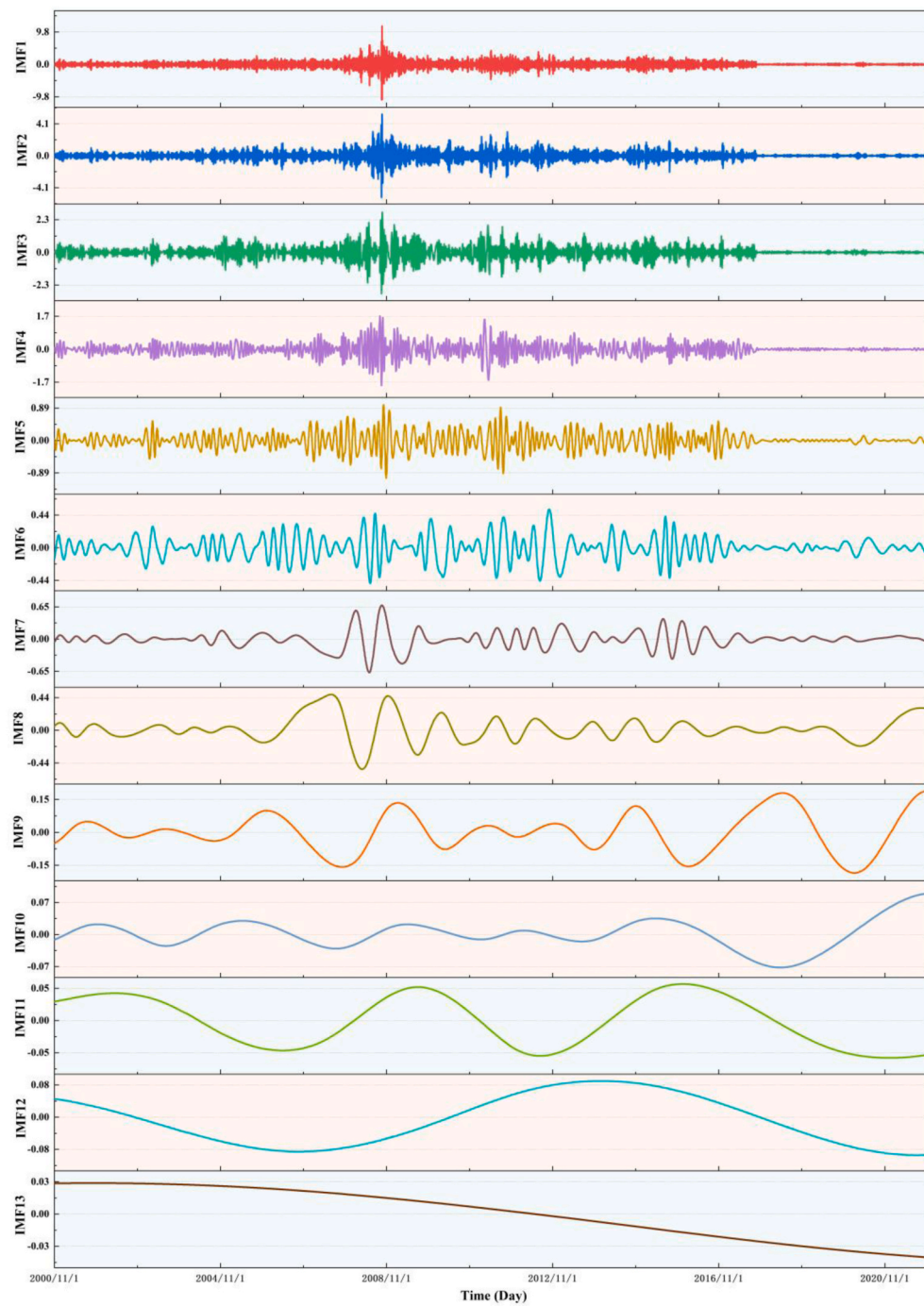


Fig. 8. Error series decomposition results of WTI daily crude oil futures prices.

can effectively improve the efficiency of proof parameter selection (Wu et al., 2020, 2023; Wang et al., 2023; Zhang et al., 2022; Zhu et al., 2019; Chen et al., 2023). The parameters of VMD are set as follows: the convergence tolerance is 10^{-6} . The results of the decomposition are shown in the following Figs. 4 and 5.

4.3. Crude oil futures prices forecasting results based on VILIG model

The parameters of VILIG model are set as follows in Table 2.

The initial forecast result of WTI daily crude oil futures price is shown in Fig. 6.

The initial forecast result of Brent daily crude oil futures price is shown in Fig. 7.

From Figs. 6 and 7, we can acknowledge that the upper IMF component is more nonlinear and non-stationary, and the initial prediction is relatively poor. The fit of the lower IMF component is higher. Therefore, accurate prediction of the error series has a crucial impact on improving the accuracy of the model. Subsequently, we applied ICE-EMDAN to decompose the error series in order to reduce the non-linearity and non-smoothness of the error series while improving the model accuracy. The following Fig. 8 shows the decomposition results of the WTI crude oil futures price error series.

The following Fig. 9 shows the decomposition outcomes of the Brent crude oil futures price error series.

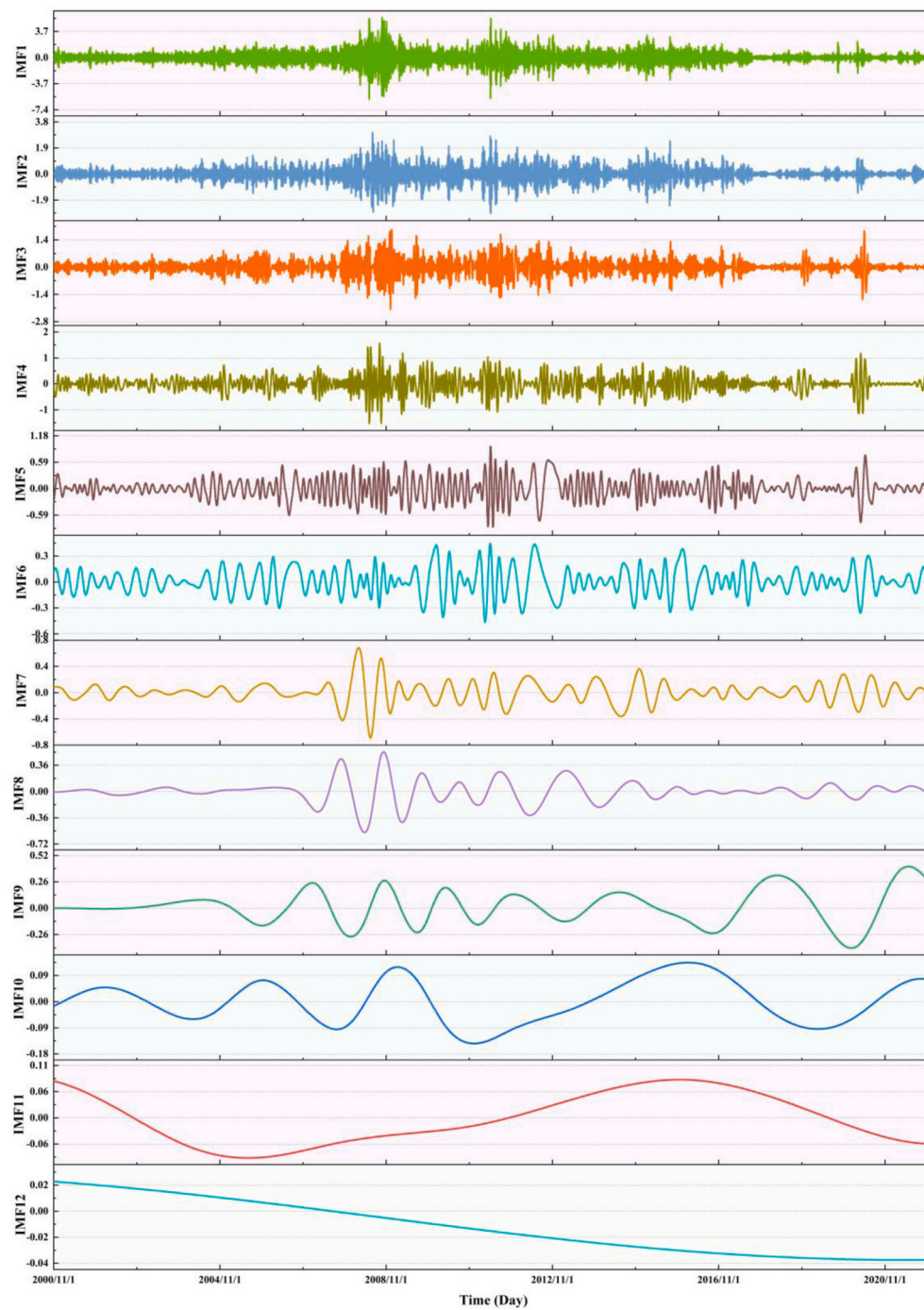


Fig. 9. Error series decomposition results of Brent daily crude oil futures prices.

4.4. Comparison of the results of various prediction models

To test the validity of the VILIG model, this study chose the 16 models for comparison. In the first stage, six deep learning models such as LSTM, GRU, Deep Neural Network (DNN) and Echo State Network (ESN) were selected to verify the overall performance of LSTM and GRU respectively. Subsequently, in the second stage, we combine signal decomposition techniques such as VMD, Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN), EEMD with deep learning techniques to find the best performing combined model. In the third stage, two models, VMD-LSTM-ICEEMDAN-GRU and VMD-GJO-LSTM-ICEEMDAN-GRU, were used to validate the effectiveness of the IGJO algorithm. In the fourth stage, VMD-DNN (Liu et al., 2021) and

ICEEMDAN-PE-EMD-CSSA-KELM (Sun et al., 2022) model were selected to valid the effectiveness of the VILIG model. Table 3 shows the abbreviations of each model.

Table 4 shows the parameters of the other models.

The following Table 5 shows the evaluation indicators of each model on the WTI crude oil futures price data set.

The following Table 6 shows the evaluation indicators of each model on the Brent crude oil futures price data set.

5. Discussion

Although statistical models can effectively predict crude oil futures prices, they require a high quality of data and need to meet certain

Table 3

The abbreviations of each model.

Model	Abbreviation
VMD-LSTM	M1
VMD-GRU	M2
ICEEMDAN-LSTM	M3
ICEEMDAN-GRU	M4
CEEMDAN-LSTM	M5
CEEMDAN-GRU	M6
EEMD-LSTM	M7
EEMD-GRU	M8
VMD-LSTM-ICEEMDAN-GRU	M9
VMD-GJO-LSTM-ICEEMDAN-GRU	M10
VMD-DNN	M11
ICEEMDAN-PE-EMD-CSSA-KELM	M12

Table 4

The parameters of each model.

Model	Parameter
LSTM/GRU	Set the maximum number of iterations to 100. Set learning rate to 0.005. Set the number of hidden layer neurons to 200.
EEMD/CEEMDAN/ ICEEMDAN	The time domain frequency domain toolbox of MATLAB version 2021a is used to perform the decomposition. Set the maximum number of iterations to 600. Set the ratio of adding white noise to 0.05.

Table 5

Comparison of various models in prediction performance on the WTI crude oil futures price data set.

Model	MSE	RMSE	MAE	MAPE	R ²
VILIG	0.13	0.36	0.28	0.01	0.999
GRU	92.19	9.60	7.12	0.16	0.580
LSTM	83.96	9.16	7.35	0.16	0.552
DNN	106.21	10.31	9.53	0.17	0.496
ESN	184.69	13.59	11.11	0.23	0.114
ELM	135.58	11.64	8.95	0.21	0.209
GRNN	210.78	14.52	10.16	0.19	0.124
M1	25.90	5.09	4.55	0.09	0.978
M2	31.48	5.61	4.71	0.09	0.854
M3	30.79	5.55	4.50	0.10	0.861
M4	26.32	5.13	3.87	0.10	0.967
M5	54.54	7.39	5.51	0.13	0.849
M6	58.80	7.67	7.16	0.13	0.842
M7	50.39	7.10	6.04	0.12	0.852
M8	71.03	8.43	7.57	0.15	0.725
M9	3.64	1.91	1.35	0.03	0.985
M10	2.58	1.61	1.18	0.03	0.986
M11	36.32	6.03	4.32	0.10	0.855
M12	1.20	1.10	0.80	0.02	0.995

conditions, so the accuracy of the prediction is relatively low. To compensate the shortcomings of statistical models, in recent years, some scholars have introduced intelligent algorithms such as neural network methods, support vector machines and intelligent optimization algorithm to crude oil futures price prediction. These methods have strong non-linear mapping, and high computational efficiency, which can precisely predict changes in crude oil futures prices. The model is important in preventing risks in the international crude oil market, monitoring changes in the international economic environment and anticipating possible changes in crude oil prices.

5.1. WTI crude oil futures price data set

Fig. 10 presents the bubble plots of *RMSE* and *MAE* for VILIG and each single model on the WTI crude oil futures price test set.

From Table 5 and Fig. 10, the *RMSE* and *MAE* of the VILIG model on

Table 6

Comparison of various models in prediction performance on the Brent crude oil futures price data set.

Model	MSE	RMSE	MAE	MAPE	R ²
VILIG	0.57	0.75	0.51	0.01	0.997
GRU	91.05	9.54	7.35	0.13	0.526
LSTM	82.59	9.09	7.28	0.14	0.585
DNN	96.48	9.82	7.98	0.14	0.502
ESN	144.84	12.03	9.37	0.17	0.236
ELM	110.74	10.52	8.08	0.15	0.407
GRNN	190.40	13.80	11.19	0.20	0.038
M1	26.97	5.19	3.91	0.07	0.868
M2	34.00	5.83	4.61	0.08	0.812
M3	42.54	6.52	5.39	0.10	0.839
M4	33.70	5.81	4.45	0.09	0.847
M5	39.73	6.30	5.00	0.09	0.802
M6	36.47	6.04	4.54	0.09	0.825
M7	52.54	7.25	5.85	0.11	0.811
M8	40.24	6.34	5.27	0.10	0.812
M9	7.18	2.68	2.12	0.04	0.973
M10	3.19	1.79	1.27	0.02	0.982
M11	38.48	6.20	5.21	0.10	0.854
M12	4.95	2.22	1.81	0.03	0.985

the test set of WTI crude oil futures prices are 0.36 and 0.28, respectively. Compared with single models such as LSTM, DNN, there is a substantial decrease in each error metric and a substantial improvement in *R²*, indicating that the deviation between the predicted and actual sequences of the VILIG model is minimal. In addition, among single models, the *RMSE* of LSTM is 9.09, Compared with other single models, the LSTM model decreased in *RMSE* by 4.71%, 7.43%, 24.44%, 13.59% and 34.13%, respectively. The reason can be attributed that LSTM adds input gates, output gates and forgetting gates to the traditional recurrent neural network, which can store the information of long sequences more effectively and tackle the problem that the gradient disappears during the processing of long-time sequences by RNN.

Fig. 11 presents the radar plots of *RMSE* and *MAE* for VILIG and each hybrid model on the WTI crude oil futures price test set.

From Fig. 11, compared with single deep learning models, models based on the idea of "decomposition-prediction-ensemble" has a significant reduction in *MSE*, *RMSE* and other evaluation metrics. Take M1 and LSTM as examples, although the LSTM model outperformed those single models, the *RMSE* and *MAE* of M1 model are 5.19 and 3.91 respectively, decreasing 42.9% and 46.29% respectively. The reason can be attributed that the introduction of signal techniques can further extract the characteristics of the original price series while removing the effect of noise in the dataset, thus making it substantially less difficult to accurately predict crude oil futures price series with strong non-linearities. Moreover, among the models with "single decomposition-prediction-ensemble" idea, the VMD-LSTM and ICEEMDAN-GRU models have the highest accuracy. The reason can be attributed that the residual noise that may exist in the decomposition process of EEMD algorithm, and the pseudo-modality that exists in the decomposition process of CEEMDAN algorithm, which has a negative impact on the final prediction results. In addition, compared with the model based on the "single decomposition-prediction-ensemble" idea, the model based on the secondary decomposition idea has a further improvement in accuracy. Take M1 and M9 as example, the *RMSE* and *MAE* of M10 is 1.79 and 1.27, which has reduced 62.48% and 70.33% respectively. The reason can be attributed to that the introduction of secondary decomposition technique can further decompose the sub-models that still have strong nonlinearity after preliminary decomposition, thus reducing the difficulty of accurate prediction of complex sub-models.

5.2. Brent crude oil futures price data set

Fig. 12 presents the radar plots of *RMSE* and *MAE* for VILIG and each single model on the Brent crude oil futures price test set.

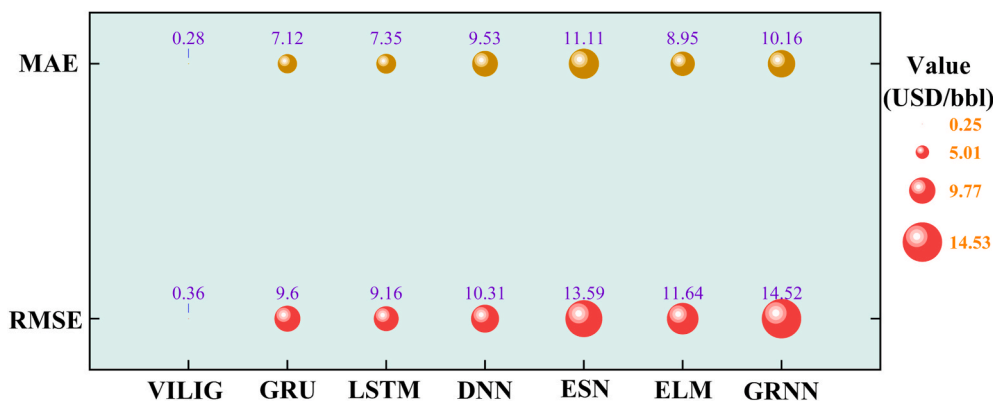


Fig. 10. Bubble plots of RMSE and MAE for VILIG and each single model on the WTI crude oil futures price test set.

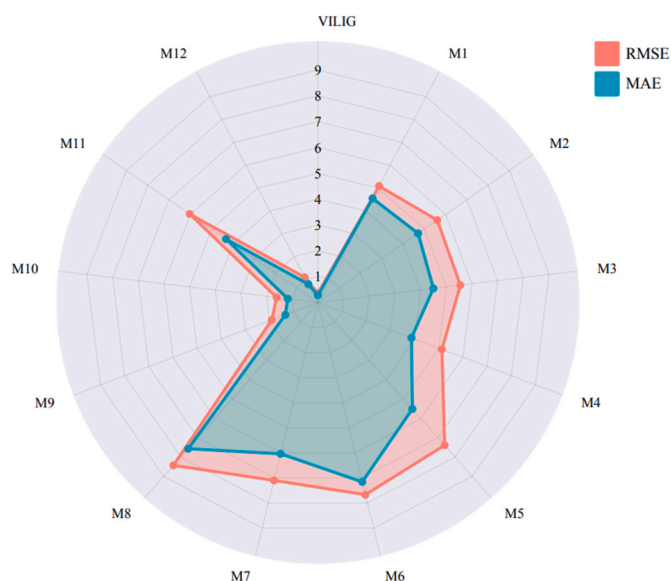


Fig. 11. Radar plots of RMSE and MAE for VILIG and each hybrid model on the WTI crude oil futures price test set.

From Fig. 12, on the WTI crude oil futures price test set, the *RMSE* of LSTM and GRU models are 9.09 and 9.54, respectively, which are much lower than DNN, GRNN and other models. Hence, in this study, we selected the LSTM and GRU models to predict the crude oil futures price series.

Fig. 13 presents the heat map of RMSE and MAE for VILIG and each hybrid model on the Brent crude oil futures price test set.

From Fig. 13, the *RMSE* of VMD-LSTM and ICEEMDAN-GRU models are 5.19 and 5.81 respectively, which are much lower than CEEMDAN-LSTM and ICEEMDAN-LSTM model in the "single decomposition-prediction-ensemble" idea. Consequently, we introduced VMD to perform the preliminary decomposition of the raw crude oil futures price series and LSTM to predict each sub-model separately. In addition, the introduction of an intelligent optimization algorithm to optimize the hyper-parameters of the LSTM can further improve the accuracy of the model. Take VILIG, M9 and M10 as example, compared with M9, the introduction of the original GJO algorithm to optimize the hyper-parameters of the LSTM contributed to a 33.21% and 40.09% decrease in the *RMSE* and *MAE* of M10 on the Brent test set, respectively. However, the final results of the model are affected by the fact that the original GJO algorithm tends to fall into local optimum. After adding strategies such as Tent mapping to further improve the GJO algorithm, the *RMSE* and *MAE* of the VILIG model decreased by 58.1% and 59.84%,

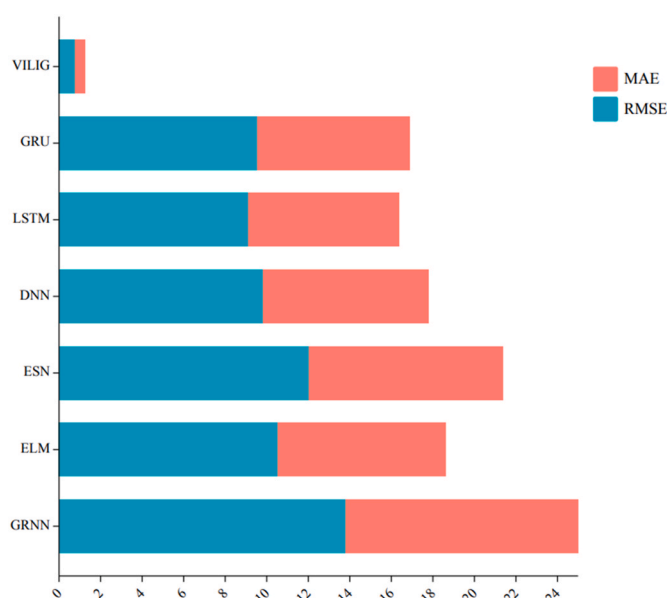


Fig. 12. Stacked histograms of RMSE and MAE for VILIG and each single model on the Brent crude oil futures price test set.

respectively. Therefore, it is effective to use the IGJO algorithm to optimize the hyperparameters of the LSTM.

5.3. Uncertainty estimation

Although the VILIG model achieves good accuracy on the test sets of WTI and Brent crude oil futures prices, crude oil futures prices are closely related to political, economic and other factors, and have a large amount of uncertainty. The interval estimation of the results of the deterministic point forecast results can provide significant guidance for risk assessment and uncertainty assessment. The kernel density estimation method is an interval analysis method often used in the fields of load forecasting, wind forecasting, etc. It does not require any additional assumptions on the sample distribution. The KDE estimation results of the error series of the VILIG model on the test set of WTI and Brent are shown in Fig. 14.

Subsequently, we perform interval estimation of the results of the point prediction at confidence intervals of 95%, 90%, and use *PICP* to evaluate the model performance. The results are shown in Table 7.

From Fig. 14., the *PICP* of the VILIG model on the WTI test set is 0.967 at the 95% confidence level, much higher than at the 90% and 85% confidence levels, indicating the validity of interval estimation.

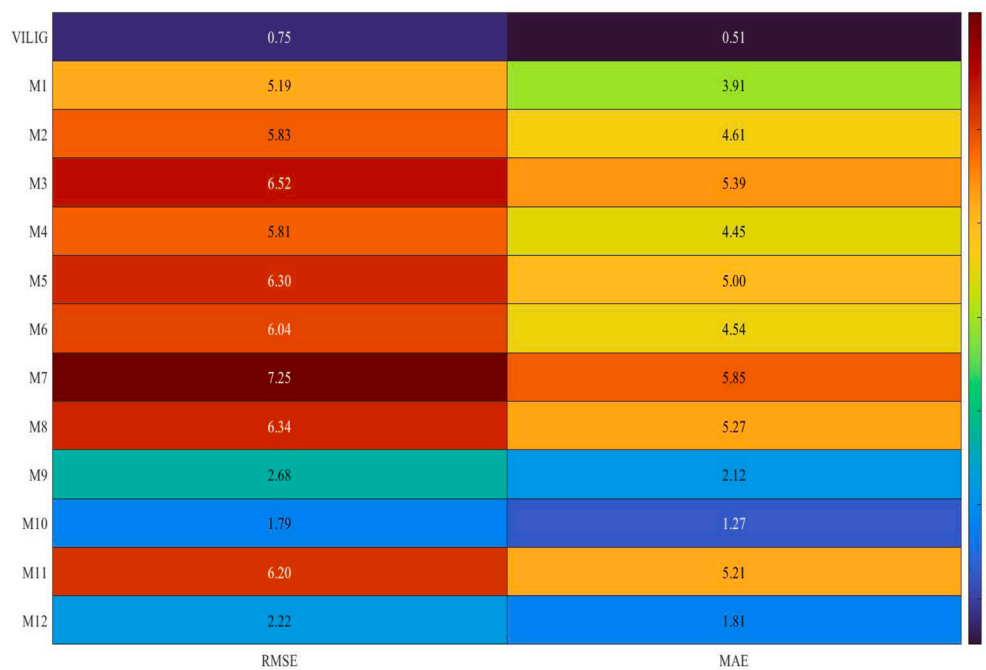


Fig. 13. Heat map of evaluation metrics for each model on the Brent crude oil futures oil price test set.

5.4. Model validity analysis

The combined VILIG model developed in this study achieves excellent performance in both point prediction and interval prediction relative to other models, and its inevitability is analyzed from the perspective of modeling ideas, mainly including: (1) The secondary modal decomposition can dig deeper into the characteristic information of the price time series, which makes the individual modal predictions more and better targeted. (2) IGJO algorithm makes the LSTM hyperparametric optimization more objective and effective, overcoming the subjective arbitrariness of previous parameter settings, and further improving the forecasting effect. (3) The error correction model tracks the white noise characteristics of crude oil futures price changes and incorporates them into the overall forecasting framework, which further reduces the forecasting error.

5.5. Model deficiencies

Although the VILIG model has a substantial improvement in accuracy, which is superior and competitive, we can improve it from the following ways.

- (1) The improved golden jackal algorithm can also be combined with other optimization algorithms (such as snake algorithm, etc.).
- (2) The model proposed in this paper still suffers from some errors in accuracy. There are several main reasons for it. The crude oil futures price is closely linked to many factors such as the political and emergency events, the equity index and the consumer price index. Therefore, in the subsequent study, we will gradually take these factors into account to improve the precision of forecasting crude oil futures prices.

6. Conclusion and outlook

As an important energy source, the price trend of crude oil has a vital influence on the economy and society. To tackle the non-linear characteristics of crude oil futures prices, this research proposes a novel hybrid model VILIG for crude oil futures price predicting and compares it with ICEEMDAN-PE-EMD-CSSA-KELM model and other 17 predictive

models. Ultimately, we introduced KDE to perform interval estimation based on the point prediction results. The outcome showed that the VILIG model outperforms other 18 models in all indicators. Moreover, compared with VMD-GJO-LSTM-ICEEMDAN-GRU model, the *RMSE* of the VILIG model has reduced 81.15% and 58.10% respectively. In addition, the *PICP* of the VILIG model was 0.967 at the 95% confidence level, much higher than at the 90% and 85% confidence levels. The main innovations of this study are: (1) the model framework of "decomposition-prediction-ensemble-error correction" is established in this study, which enables the modal decomposition to explore the characteristic information of price series, the sub-module prediction to be more targeted, and the error correction to further enhance the prediction effect. (2) The use of targeted secondary modal decomposition for different characteristic series makes the characteristic information more visible. (3) IGJO algorithm for LSTM hyperparameter search to make it more objective and reasonable.

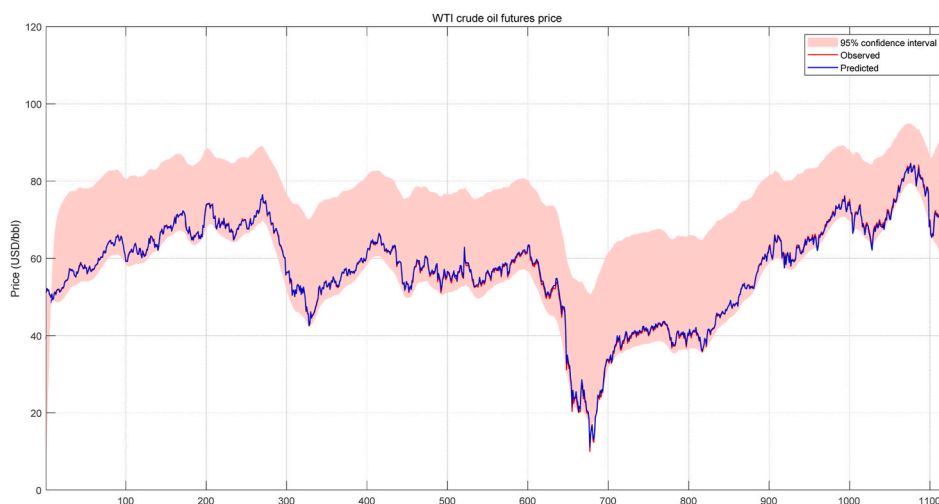
However, the VILIG model can be improved from the following ways. (1) we can combine improved golden jackal algorithm with other optimization algorithms to see if there is any improvement in the precision of the model. (2) factors such as the US dollar index, stock index, etc. which are closely linked to crude oil futures prices can be considered for inclusion. We hope that later research can further improve crude oil futures prices forecasting model.

CRediT authorship contribution statement

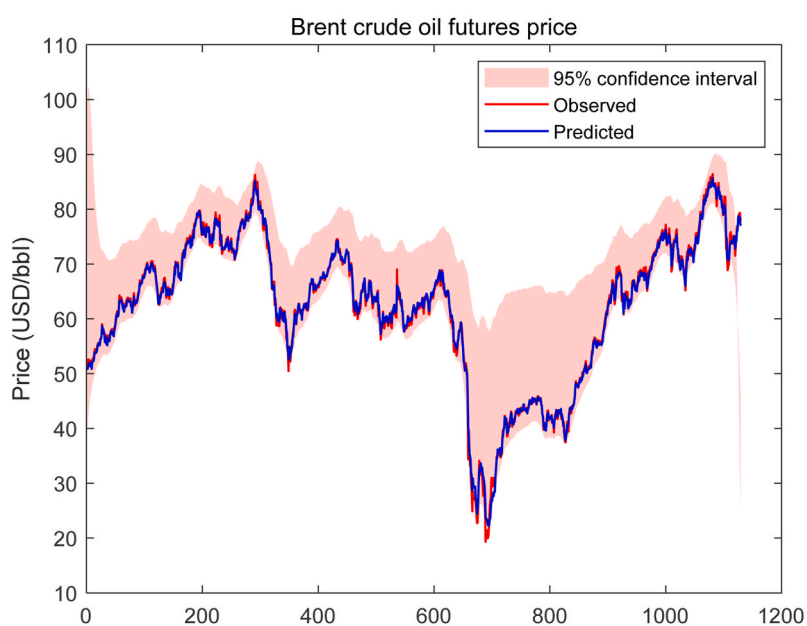
Junhao Wu: Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Jinghan Dong:** Experiments conduction. **Zhaocai Wang:** Writing – review & editing, Writing – original draft, Project administration, Methodology. **Yuan Hu:** Experiments conduction. **Wanting Dou:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.



(a) WTI interval estimation results at 95% confidence level



(b) Brent interval estimation results at 95% confidence level

Fig. 14. Results of interval estimation of error series.

Table 7

Interval estimation results at different confidence levels.

Market	Confidence level	PICP	PINAW
WTI	95%	0.97	47.45%
	90%	0.93	16.42%
	85%	0.87	14.39%
Brent	95%	0.96	31.11%
	90%	0.93	40.25%
	85%	0.89	33.55%

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